1 DIGITAL DEVICES

Digital devices are pieces of hardware that use computers or microcontrollers, and they are found everywhere in our digital world. They enhance and support how we live our lives every day. They can connect and work together to give us the data we need, when and where we need it.

Digital devices are always developing. This changes the way in which they are used by individuals, organisations and local, national and global societies.



LEARNING OBJECTIVES

- Be aware that mainframe computers are used for complex processing tasks and microprocessors are embedded in products such as washing machines
- Understand that laptop and desktop computers are types of personal computers. Some laptops are used as desktop replacements
- Know about types of mobile phones; smartphones and specialist phones and how they connect to the network (SIM)
- Know about tablet devices
- Be able to describe the purpose and use of other digital devices such as:
  - cameras and camcorders
  - games consoles
  - home entertainment systems
  - media players
- Know about navigation aids and how they are used
- Understand the terms 'multifunctional' (e.g. mobile phones that include a camera, have limited game playing functionality and GPS) and 'convergence' (e.g. functionality of smartphones and tablet devices becomes more similar) in the context of digital devices
- Understand features of digital devices: portability, performance, storage, user interface, connectivity, media support, energy consumption, expansion capability, security features
- Be able to discuss the features of identified digital devices
- Be able to identify digital devices and associated peripheral devices that meet particular needs, including accessibility
- Know about types of output peripheral such as monitor (screen size, resolution), printer (laser, inkjet, 3D), plotter, data projector, speaker, control device and when they would be used
- Know about types of input peripheral such as keyboard, mouse, tracker ball, joystick, graphics tablet, scanner, digital camera, webcam, microphone, touch screen, OMR reader, OCR reader, bar code scanner, biometric scanner, magnetic stripe reader, chip and pin, sensor and when they would be used
- Be able to differentiate between storage devices and the media used to store data
- Know the characteristics of hard disk drives (HDD), solid state drives (SSD), optical disk drives, flash memory drives
- Know that storage devices can be internal or external
- Know about types of storage media such as hard disks, optical disks (CD, DVD, Blu-ray), flash memory devices, magnetic tape
- Know that storage media can be recordable / write once (R) and rewritable (R/W)
- Understand the terms describing the capacity of storage such as bit, byte and multiples of these (kbytes, mbytes, gbytes, tbytes) (using 1KB = 1000 bytes)

TYPES OF DIGITAL DEVICES

SUBJECT VOCABULARY

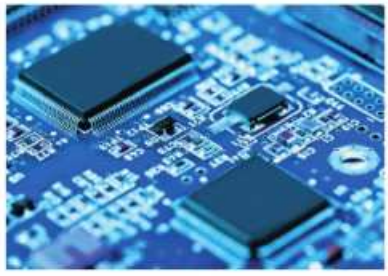
mainframe computer a large, powerful computer that can do a lot of complicated jobs quickly and can be used by a lot of people at the same time

microprocessor the device that controls what a computer does; it takes data as input, does something with it and provides output

There are many types of digital devices. They range from very powerful **mainframe computers**, used by large organisations for complex processing tasks such as statistical analysis and bulk data processing, to **microprocessors** used to control washing machines, televisions and other household appliances. Examples of the devices you will consider include personal computers, digital cameras and home entertainment systems.



▲ **Figure 1.1** A mainframe computer



▲ **Figure 1.2** A microprocessor

SKILLS CRITICAL THINKING
INTERPRETATION
COMMUNICATION

ACTIVITY

▼ DIGITAL DEVICES

1 Research the difference between a mainframe computer, a server and a supercomputer.

2 Create a list of the digital devices in your home that use microprocessors.

PERSONAL COMPUTERS

Personal computers (PCs) are common in homes and offices. They come in many different shapes and sizes, such as desktops and laptops.

SUBJECT VOCABULARY

upgrade to make a computer better and able to do more things

peripheral device equipment that is connected to a computer and used with it

DESKTOPS

Desktops have more space for components than laptops and often provide users with the option to **upgrade** them or add additional components. A desktop computer usually needs to have **peripheral devices** connected to it, such as a monitor, a printer, a mouse and a keyboard. For more information about peripheral devices, see pages 19, 20 and 32.



▲ Figure 1.3 A desktop computer

Some desktops are 'all-in-one'. This means that they combine the monitor with the PC hardware, as shown in Figure 1.4.



▲ Figure 1.4 An all-in-one computer

LAPTOPS

GENERAL VOCABULARY

portable easy to carry or move

Laptops include a keyboard, a screen, a track pad (see page 23) and a rechargeable battery. Having these features means that they can be taken away from the desk, which makes them an example of a **portable** device.



▲ Figure 1.5 A laptop computer

Some laptops are called 'desktop replacements'. These tend to be larger than other laptops, as well as having a bigger and better-quality screen and higher performance levels.

SUBJECT VOCABULARY

embedded computing computing hardware that is fixed into position and carries out a specialist task
physical computing interactive systems that can sense and respond to the world around them

SINGLE-BOARD COMPUTERS

Single-board computers (SBCs) are affordable computers used in education, **embedded computing** projects and **physical computing** projects. The Raspberry Pi Zero (see Figure 1.6) is an example of an SBC. The unit itself costs very little and uses a cheap microSD card as its storage.



▲ **Figure 1.6** The Raspberry Pi Zero (top) and its microSD card (top left)

MOBILE PHONES

Mobile phones use a SIM card to connect to a mobile phone network. SIM stands for subscriber identity module, and a SIM card is used to identify the subscriber to a mobile phone network.

DID YOU KNOW?

Sri Lanka has more mobile phones than people. In 2015, for every 100 Sri Lankan citizens there were 113 mobile phones.



▲ **Figure 1.7** The SIM card from a mobile phone

SUBJECT VOCABULARY

user the person who uses a computer system

SPECIALIST PHONES

Some mobile phones have specialist features to provide **users** with functions that meet particular user needs. For example, some phones have an emergency button that is linked to a list of emergency contacts. When this button is pressed, the phone will call each person on the list until someone answers.

SKILLS CRITICAL THINKING
PERSONAL AND SOCIAL RESPONSIBILITY
COMMUNICATION
INTERPERSONAL SKILLS

ACTIVITY

▼ EMERGENCY BUTTON

Discuss which groups of people could need a phone with an emergency button. How would this phone meet their needs?

GENERAL VOCABULARY

accessibility a measure of how easy something is to use, especially by people with disabilities

Other phones meet users' **accessibility** needs. Examples include the Alto 2 'talking phone', shown in Figure 1.8, which is a specialist mobile phone for blind and partially sighted people. It meets its users' needs because every feature and function is spoken aloud. Other phones provide fewer, larger buttons and connections for hearing aids.

SUBJECT VOCABULARY

Wi-Fi (wireless fidelity) a way of connecting computers or other electronic machines to a network or the internet by using radio signals rather than wires
games console an electronic device that is used for playing computerised video games on a screen
application a program that allows a user to perform a task; applications are often called apps
virtual keyboard a keyboard displayed on and used via a touch screen



▲ **Figure 1.10** A virtual keyboard



▲ **Figure 1.8** The Alto 2 'talking phone' reads the names of contacts and text messages aloud to its user



▲ **Figure 1.9** Some mobile phones use tactile keypads, which are more responsive than other keypads, and give the user a choice of on-screen text size and colours

SMARTPHONES

Smartphones are small computers with **Wi-Fi** and mobile phone connectivity to allow them to make phone calls and access the internet. They also include features of other devices such as cameras, media players and hand-held **games consoles**. They have a more advanced operating system than other mobile phones. **Applications** (or apps) can be downloaded onto the smartphone, which allow users to customise their smartphones with entertainment, educational and business features. Most smartphones use a touch screen to allow users to input information. A **virtual keyboard** is used to enter text, numbers and other characters.

Because they combine so many features, smartphones use more power than other types of mobile phone. This means that they have a shorter battery life and need to be charged more regularly than other mobile phones.

SKILLS CRITICAL THINKING
INTERPRETATION
COMMUNICATION

ACTIVITY

▼ SMARTPHONES AND MOBILE PHONES

1 Research which smartphone features use the most power.
2 Make a list of activities for which a standard mobile phone would be more suitable than a smartphone.

SKILLS CRITICAL THINKING
PERSONAL AND SOCIAL RESPONSIBILITY

ACTIVITY

▼ ACCESSIBILITY

The Royal National Institute of Blind People is a UK organisation that supports people with sight impairments. Do an internet search for the key terms 'RNIB mobile phone accessibility' to visit the RNIB's website, which will give you lots of information about the accessibility features of mobile phones and smartphones.

Watch the video on the RNIB's website about the accessibility features of major smartphones and make a list of the features mentioned. Which features did you already know about?

TABLETS

Tablet devices or tablets are bigger than smartphones, but have similar features. For example, a tablet device has a touch screen, apps and Wi-Fi connectivity to provide access to the internet. Some tablet devices have SIM card slots to allow internet connectivity using the mobile phone network, so that they can be online when they are not within range of a Wi-Fi signal.



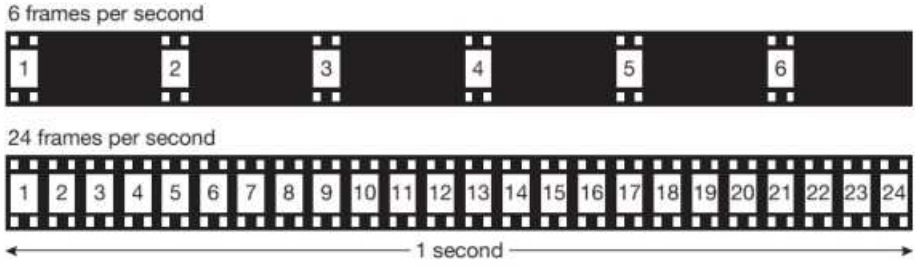
▲ **Figure 1.11** A tablet device

CAMERAS AND CAMCORDERS

DID YOU KNOW?

Moving images (movies) are simply a sequence of images called 'frames'. Figure 1.12 shows how the frame rate affects the quality of a movie.

Digital cameras and camcorders use light sensors to capture images formed by light passing through the device's lens. Traditionally, cameras are used to capture still images and camcorders are used to capture moving images. However, most digital cameras can now film moving images and most camcorders can photograph still images.



▲ **Figure 1.12** High frame rates produce smooth results, whereas low frame rates can produce results that appear to stutter or jump

GENERAL VOCABULARY

compensate to reduce or correct a negative effect

SUBJECT VOCABULARY

pixel a small dot that helps to make up an image

The quality of the image captured by the camera depends on the quality of the lens, the image processor and the resolution of the sensor.

- A good lens allows light to travel through it without introducing any defects. It also allows the user to choose how much light can travel through it, which affects the final image.
- A good image processor can **compensate** for poor lighting conditions.
- Digital images are made up of small dots called **pixels**. Better-quality sensors can capture more detail and produce images with a greater number of pixels. The sensor resolution of a camera is expressed as the number of pixels that can be captured, as shown in Table 1.1. Some cameras use more than one sensor, each dedicated to a different colour or wavelength of light.

SUBJECT VOCABULARY

resolution the number of pixels used by a screen to display an image
pixelated an effect that creates an unclear image consisting of large individual pixels that are visible to the human eye



▲ **Figure 1.13** The area of the image with the leaf becomes **pixelated** when enlarged

▼ **Table 1.1** Maximum display size at different **resolutions**

Resolution (number of pixels)	Maximum display size of image (in pixels, width × height) before pixelation occurs
1.3 megapixel (1.3 million pixels)	1280 × 960
2 megapixel	1600 × 1200
3 megapixel	2048 × 1536
4 megapixel	2272 × 1704
8 megapixel and above	2560 × 1920

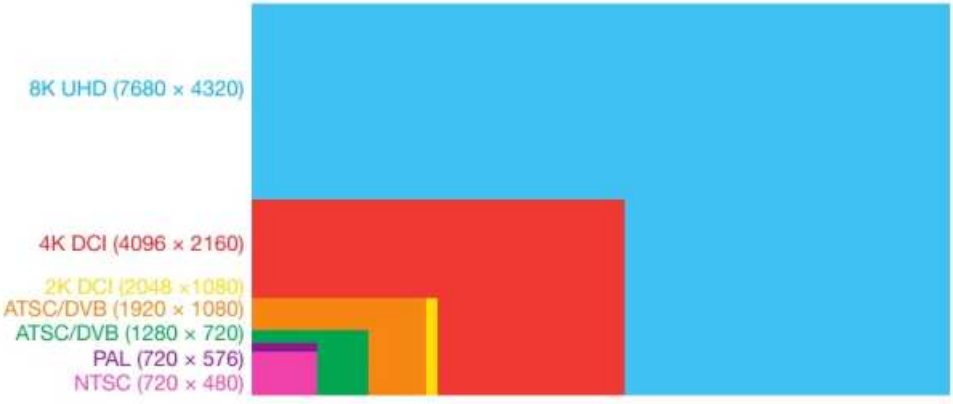
HOME ENTERTAINMENT SYSTEMS

TELEVISIONS

Televisions display still and moving images on a screen. The quality of the image is set by the number of pixels that are used to display the image. This is referred to as the screen's resolution. High definition (HD) television screens contain a larger number of pixels, which means that they have a higher resolution than standard definition televisions.

The resolution of a television in pixels is stated as horizontal pixels × vertical pixels. Often, television manufacturers do not specify the number of horizontal pixels and instead refer only to the number of vertical pixels. For example, a resolution of 1280 × 720 is often stated as 720p and 1920 × 1080 is stated as 1080p.

Ultra High Definition (UHD) television screens are sometimes referred to as 4K or 8K because they have a horizontal resolution of approximately 4,000 or 8,000 pixels. 8K screens make each pixel impossible to tell apart even when users are close to the screen.



▲ **Figure 1.14** Comparison chart showing different screen resolutions

SKILLS CRITICAL THINKING
COMMUNICATION
SELF-DIRECTION
INTERPERSONAL SKILLS

SUBJECT VOCABULARY

stream play a file on your computer while it is being downloaded from the internet, rather than saving it as a file and then playing it
smart the ability of a device to use data from sensors to perform a programmed action independently; smart devices are often also able to connect to the internet

ACTIVITY

▼ TYPES OF TELEVISION

1 Research the cost of HD, 4K and 8K televisions.
2 Research the types of content that is broadcast or available to buy in each resolution. How easy would it be to **stream** content in each resolution?
3 Estimate how far you sit from your television when watching television at home, then research how close to the screen a user must be in order to notice the quality of the three different resolutions.
4 Discuss whether it is worth paying the extra cost for 4K and 8K televisions in order to watch television. Do you think there is any point in creating 16K screens for home use?

For the production of sound, most televisions come with built-in speakers. However, these are often small and of limited quality because the screen is usually very thin. Most modern televisions can be connected to an external sound system to improve their sound quality.

Smart televisions use apps and can download and stream content from the internet.

SOUND SYSTEMS

Sound systems can produce loud, rich sound using high-quality speakers and amplifiers. Some speakers contain built-in amplifiers.

Sound systems can play music from CDs or from local storage. They can usually be connected to personal devices like smartphones, **media players** and tablet devices using wired connections like **USB** or wireless connections like **Bluetooth**. They can also be connected to a local network via Wi-Fi to play music that is stored on connected devices. Some sound systems can also connect to the internet to play music stored online.

SUBJECT VOCABULARY

media player an electronic device that can store and play digital music and videos and show digital photographs
USB (Universal Serial Bus) a standard for wired connectivity that can also supply electric power
Bluetooth wireless connectivity that allows devices to connect over short distances
set-top box (STB) a device that sends video and audio received from a broadcaster to a television

PERSONAL VIDEO RECORDERS

A personal video recorder (PVR) is a device that records broadcasted content so that it can be watched at a later date.

Some devices fall into multiple categories. For example, some satellite television devices contain both a **set-top box (STB)** and a PVR.

BLU-RAY AND DVD PLAYERS

Blu-ray and DVD players connect to televisions in order to play films and other content that is stored on DVD or Blu-ray disks (see page 39 for more on Blu-ray media). Blu-ray players will usually play DVD disks, but DVD players will not play Blu-ray disks.

Blu-ray disks can store HD movies, which have higher-quality picture and sound. Newer 4K Ultra HD Blu-ray players can play 4K content on 4K televisions.



▲ **Figure 1.15** A Blu-ray player

GAMES CONSOLES



▲ **Figure 1.16** How many games consoles have you tried?

Games consoles are designed to enable users to play video games on a television screen. Games are provided on disks or as downloads from the internet. They use controllers, which are often wireless, to control the characters, vehicles and/or objects in the game.

SUBJECT VOCABULARY

avatar a figure that represents a person in a computer game or other virtual environment

Some consoles use motion sensors to allow the player to control the game with gestures and body movements. Other games use virtual reality controllers and headsets to immerse the player in a realistic gaming experience, where their own movements in the real world are replicated by an **avatar** in the virtual world of the game.



▲ **Figure 1.17** Have you played any virtual reality games?

SKILLS CRITICAL THINKING
PERSONAL AND SOCIAL RESPONSIBILITY
COMMUNICATION
INTERPERSONAL SKILLS

ACTIVITY

▼ VIDEO GAMES

Discuss the age ratings of popular games. Do you think these age ratings are appropriate? Do you think age ratings are necessary? Explain the reasons for your answer.

Modern games consoles are multifunctional (see page 15). They may have apps and connectivity that provide access to local networks and the internet as well as the option to play movies and music. Some consoles also have disk drives to play films and other content on DVD, Blu-ray or 4K Ultra HD Blu-ray media.



▲ **Figure 1.18** Consoles can often do more than just play games

SUBJECT VOCABULARY

data values that represent information

Handheld versions of games consoles provide mobile gaming. Handheld consoles have a built-in screen and less storage than a full-size console. This means that the games that can be played on these devices often have reduced image and sound quality, reduced game complexity and limited storylines in order to reduce the amount of **data** that needs to be stored.

MEDIA PLAYERS

DID YOU KNOW?

Network attached storage (NAS) devices often have media streaming capabilities built in.

Media players provide video and audio content to a television. This content could be streamed directly from the internet or accessed from networked or local storage. Media players can be connected to a television directly or using a wired or wireless network. They are controlled using a remote control or, in some cases, a smartphone app. Some media players offer output in 4K.



▲ **Figure 1.19** Have you used any other examples of media players?

SKILLS CRITICAL THINKING
COMMUNICATION
INTERPERSONAL SKILLS

DID YOU KNOW?

A Raspberry Pi Zero single-board computer and free software can be used as a media player. This makes it a very affordable alternative to a television and media player, or a smart television.

SUBJECT VOCABULARY

flash memory a form of storage that stores data as electrical charges held in tiny electrical cells

ACTIVITY

▼ SMART TELEVISIONS

Smart televisions have the streaming and networking functions of a media player built in. Do you think that smart televisions are going to replace media players? Discuss your reasons for thinking that this may or may not happen.

Personal media players are compact, portable devices with local storage to hold media files for playback. They are useful when travelling or exercising due to their small size and long battery life. They often use solid state **flash memory** storage (see page 62), which is not affected by being moved or shaken.



▲ **Figure 1.20** Personal media players are portable so they need to function while on the move

NAVIGATION AIDS

Navigation aids (such as Sat-Nav) can calculate the best route between two or more locations and can provide updates to the route if it is not followed accurately. They are commonly used in cars, delivery vans and ships. They provide visual prompts and alerts to help drivers take the correct route, such as by taking a particular turning. Specialist devices with audio alerts and waterproof and shockproof cases are available for walkers, cyclists and runners.

Navigation aids use information from **GPS** satellites to determine the exact location of the device on Earth. Navigation aids also use orientation sensors to know which way the device is currently pointing, and the device will display the user's position and orientation on a map. Maps are either stored permanently on the device or downloaded from the internet when they are needed.

GPS does not need internet connectivity in order to work. However, navigation aids need internet connectivity in order to update map data.

Smartphones are now replacing specialist in-car navigation aids. There are many navigation apps for smartphones which have a range of features.

SUBJECT VOCABULARY

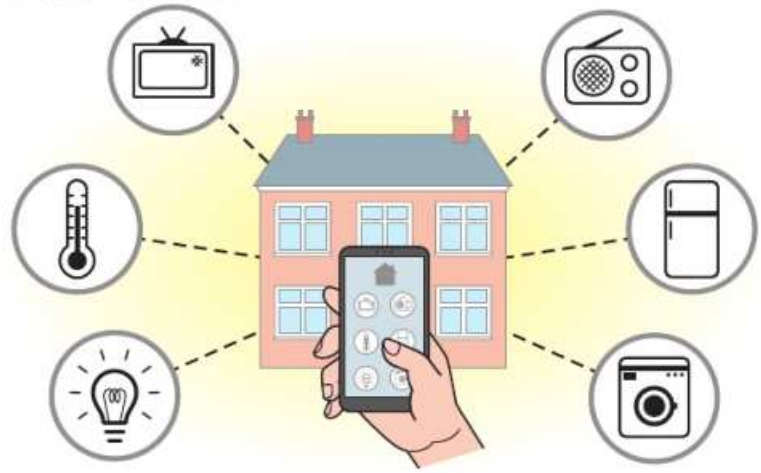
GPS the Global Positioning System that uses radio signals from satellites to show your exact position on the Earth on a special piece of equipment

DID YOU KNOW?

GPS satellites orbit more than 20km above Earth. They are accurate to within 5 metres.

HOME AUTOMATION DEVICES AND SMART ASSISTANTS

One type of emerging technology is a group of devices that can be used to create smart homes. For example, home automation devices can connect a range of digital devices which sense and control functions in the home, such as temperature and lighting. These functions can be controlled from apps on smartphones or the internet.



▲ **Figure 1.21** Home automation devices can connect digital devices to control functions in the home

MULTIFUNCTIONAL DEVICES AND CONVERGENCE

SUBJECT VOCABULARY

multifunctional device a device that can perform a range of different functions

SKILLS SELF-DIRECTION

A smartphone is a type of mobile phone, which means that its primary function is to make phone calls. It can also be used to take photographs or to function as a navigation aid, fitness tracker, music player or handheld games console. Because a smartphone can perform such a range of different functions, it is classed as a **multifunctional device**.

ACTIVITY

▼ MULTIFUNCTIONAL DEVICES

Identify another example of a multifunctional device. What is its primary function? What else can it do?

SUBJECT VOCABULARY

convergence when the designs of devices change so that they become similar to one another

As they develop, devices like smartphones often adopt technologies and features from other types of device. This is called **convergence**.

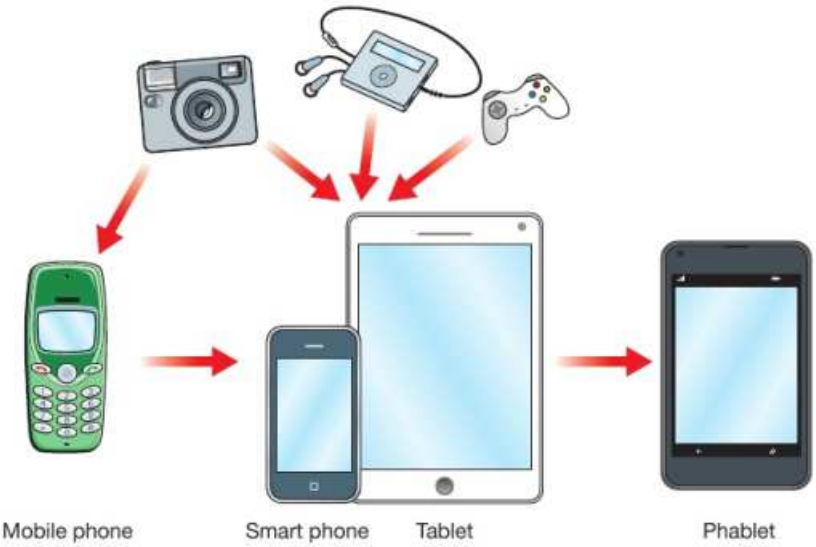


▲ **Figure 1.22** Some laptops now have touch screens that can flip 360 degrees, allowing them to operate as tablet devices



▲ **Figure 1.23** Some tablet devices come with attachable keyboards, allowing them to operate as laptop computers

Convergence blurs the distinction between different types of device. For example, smartphones and tablet devices are very similar. They are different in that a smartphone has mobile phone network connectivity and is smaller than a tablet device. However, newer smartphones are getting bigger to look more like tablets (and are known as 'phablets'), while tablet devices can now connect to mobile phone networks. Similarly, most smartphones have digital cameras, while some digital cameras have Wi-Fi and mobile phone network connectivity.



▲ **Figure 1.24** As devices develop, their features often converge

SKILLS SELF-DIRECTION

ACTIVITY

▼ CONVERGING DEVICES

Research the differences and similarities between smartphones, tablet devices and touch screen laptops.

FEATURES OF DIGITAL DEVICES

SUBJECT VOCABULARY

connectivity a device's ability to connect to networks and other devices

The digital devices that you use have a number of features, such as portability, performance and **connectivity**. Different devices will have different features.

PORTABILITY

For a device to be portable, it needs to be easy to carry and move around. This means that the portability of a device is directly related to its size and weight. For some devices, such as a television or a desktop computer, portability is unlikely to be a priority.

PERFORMANCE

SUBJECT VOCABULARY

processor one or more Central Processing Units (CPUs) that carry out software instructions
RAM Random Access Memory, which is the memory in a computer system that is used for running software
virtual memory storage used by the processor once the space in RAM has run out

A high-performance device performs its job or tasks quickly. The speed at which a device performs is determined by the speed at which it can carry out instructions from its software. Software instructions are carried out by the **processor**, which means that a faster processor will increase performance (see page 62).

Instructions are loaded into the processor from **RAM**, which means that faster RAM will also allow instructions to be loaded into the processor more quickly. RAM holds all software instructions to be carried out, so more RAM will allow the system to have more programs running at the same time. In addition, because instructions are loaded into RAM from storage, such as a hard disk (see page 39), faster storage will enable faster loading of instructions to RAM.

When the space in RAM runs out, storage is used as an overflow, and this is known as **virtual memory** (see page 60). This means that fast storage means faster access to instructions in virtual memory.

The software itself can also be written in an efficient way that makes the most of the processor's capabilities. Inefficient software can slow down the whole system and therefore has an impact on a device's performance.

STORAGE

SUBJECT VOCABULARY

file a collection of data that represents a document, image, database or similar
program a set of instructions that are carried out by the processor

Files and **programs** are stored in storage. More available storage allows users to store more files and programs. Storage speed also affects performance, as you saw in the previous section.

USER INTERFACE

SUBJECT VOCABULARY

user interface the system that allows a user to interact with a device

Users give commands to a device through the **user interface**. There are several different types of user interface.

COMMAND LINE INTERFACE (CLI)

When using a command line interface, users enter text instructions and the computer system provides results or feedback as text. This type of interface